

WITNESS-3 Preregistration

“Cosmic Beacon” — Bell-Certified, Universe-Witnessed Authorization Nonce

Series: WITNESS (third record)

Author: Derek Adam Hone / Remnant Fieldworks Inc.

Preregistration lock date: 2026-07-18 (UTC)

Status: PREREGISTERED — locked before QPU execution

0. One-paragraph summary

WITNESS-3 binds a single ExecutionProof authorization nonce to **three independent, publicly re-verifiable physical witnesses**: (1) a **CHSH Bell-test** measured on IBM quantum hardware, (2) a pulse from the **NIST public Randomness Beacon**, and (3) a byte-exact segment of **LIGO/GWOSC open gravitational-wave data** from GW150914 — the first direct detection of gravitational waves, the merger of two black holes ~1.3 billion light-years away. The same QPU measurement that seeds the nonce is a Bell test, so the run also certifies (device-dependently, with all loopholes disclosed) that the quantum correlations violate the classical bound $|S| \leq 2$. To our knowledge this is the first authorization ProofRecord whose freshness is jointly witnessed by a Bell-inequality violation, a national metrology beacon, and real astrophysical detector data — every byte of which any third party can independently re-download and re-hash.

1. Claim (single, falsifiable)

A ProofRecord constructed with a length-prefixed **fused “cosmic” nonce** over three independent witnesses (QPU CHSH counts, NIST beacon pulse, LIGO/GWOSC strain file) can be:

- (a) verified end-to-end against all three witness sources and the provider job record (W3-C1),
- (b) detected as tampered when **any single element of any witness** is substituted (W3-C2),
- (c) rejected as a cross-context replay by the two-layer (record_hash + ARK-457) defence (W3-C3), and
- (d) accompanied by a **CHSH Bell-inequality violation** certified at $\geq 5\sigma$ above the classical bound on the tested backend (W3-C4),

under the tested construction, circuit, backend, qubits, calibration, and shot count.

2. Non-Claims (explicit — these protect the credibility of the claim)

- **NOT a loophole-free / device-independent Bell test.** The two qubits are neighbouring transmons on a single chip: there is **no space-like separation** (locality / communication loophole open), readout assumes **fair sampling** (detection loophole

open), and measurement settings are fixed by the harness (freedom-of-choice loophole open). A violation here is consistent with non-classical correlations on the tested device under no-signalling and fair-sampling assumptions. It is **not** a device-independent randomness certification in the loophole-free sense.

- **Does NOT detect gravitational waves and makes NO cosmological claim.** The LIGO/GWOSC strain file is used purely as a **public, re-verifiable provenance witness** — real, published detector bytes that anyone can re-download and hash. WITNESS-3 does not re-derive any astrophysical parameter.
- **Does NOT re-run NIST's signature chain.** The beacon pulse's signature and certificate id are recorded so any auditor can verify them against NIST's published certificate.
- **Does NOT prove quantum randomness superiority, universal security, production readiness, or RF-100 conformance.** Results apply only within the tested conditions.
- `record_hash` is a **field-integrity seal, not a context-binding mechanism** (see §7, W3-C3(b)); context enforcement is delegated to the ARK-457 authorization layer.
- The fused nonce is a **provenance / freshness construction**, not a claim of information-theoretic min-entropy from any single witness.

3. CHSH Circuit & Measurement Specification

- **State:** Bell pair $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ via H on Alice + CNOT(Alice→Bob).
- **Qubits:** 2 logical (Alice = q0, Bob = q1); physical mapping by transpiler.
- **Settings (preregistered optimal angles for $|\Phi^+\rangle$, where $E(\alpha, \beta) = \cos(\alpha - \beta)$):**
 - Alice: $a = 0$, $a' = \pi/2$
 - Bob: $b = \pi/4$, $b' = 3\pi/4$
- **Four measurement circuits** (one per setting pair), in fixed order:
`ab (+E)` , `abp (-E)` , `apb (+E)` , `apbp (+E)` .
- **Measurement realisation:** to measure $\cos(\theta)Z + \sin(\theta)X$, append `ry(-θ)` then measure in the Z basis. Sign convention validated on the Aer simulator (must reproduce $S \approx +2.83$).
- **CHSH statistic:** $S = E(a, b) - E(a, b') + E(a', b) + E(a', b')$. Classical bound $|S| \leq 2$; Tsirelson bound $|S| \leq 2\sqrt{2} \approx 2.8284$.
- **Correlator:** $E = [N(00) + N(11) - N(01) - N(10)] / N_{\text{total}}$.
- **Shots:** 2,000 per setting (8,000 total), one Batch job.
- **Channel:** `ibm_quantum_platform` . **Execution mode:** Batch.
- **Backend selection:** operational, non-simulator, ≥ 2 qubits, open-plan accessible; minimum pending jobs among `{ibm_fez, ibm_marrakesh, ibm_kingston}`. Name, calibration snapshot, timestamps, job id, and provider instance recorded in `results/raw/` .
- **QPU budget gate:** `usage_remaining_seconds` ≥ 20 , else GATE-STOP (do not purchase time).
- **Provider/network unavailability at any fetch or verification step:** GATE-STOP, not FAIL.

4. Witness Sources (all publicly re-verifiable)

4.1 QPU witness

Raw per-setting counts from the CHSH job + job_id + 10-field deterministic calibration snapshot (schema identical to WITNESS-2 `calibration.py`).

4.2 NIST witness (`beacon_nist.py`)

One pulse from the NIST Randomness Beacon v2 (`/beacon/2.0/pulse/last`), fetched at execution time. Recorded fields: uri, version, chainIndex, pulseIndex, timeStamp, outputValue, certificateId, signatureValue (sha256 + 32-char prefix), and the sha256 of the exact raw pulse bytes.

4.3 Astro witness (`astro_gwosc.py`)

The public GWOSC strain file for GW150914 (H1, 32 s, 4096 Hz; `https://gwosc.org/s/events/GW150914/H-H1_LOSC_4_V1-1126259446-32.hdf5`). Recorded: url, file size, **SHA-256 of the exact file bytes**, event/detector/GPS/duration/sample-rate, and a deterministic digest over a fixed strain-sample window (offset 65536, 4096 samples, quantised at $1e-24$) so the record binds to the actual physical strain time-series.

5. Fused Nonce Construction (length-prefixed, 5 segments)

```
LP(x) = struct.pack('>I', len(x)) + x          (4-byte big-endian uint32 length prefix)

cosmic_nonce = SHA-256(
    LP(canonical_json(raw_counts))              # per-setting CHSH counts
    || LP(job_id)
    || LP(canonical_json(calibration_snapshot))
    || LP(canonical_json(nist_beacon_record))
    || LP(canonical_json(astro_witness_record))
)
```

`canonical_json = json.dumps(obj, sort_keys=True, separators=(',', ':'), ensure_ascii=True)`. Length prefixing prevents boundary-collision across the five segments. Result: 64-char lowercase hex. Each witness also carries an independent SHA-256 (`raw_counts_hash` , `calibration_hash` , `nist_hash` , `astro_hash`).

6. ProofRecord Schema (witness-proofrecord-3.0)

```
{
  "schema_version":    "witness-proofrecord-3.0",
  "cosmic_nonce":      "<64-hex fused nonce>",
  "job_id":            "<IBM Quantum job id>",
  "backend":           "<backend name>",
  "provider_instance": "<CRN / provider instance>",
  "calibration_hash":  "<sha256>",
  "raw_counts_hash":   "<sha256>",
  "nist_hash":         "<sha256>",
  "astro_hash":        "<sha256>",
  "chsh":              { "S", "abs_S", "sigma_S", "classical_bound",
                        "tsirelson_bound", "sigmas_above_classical", "correlators" },
  "bell_certified":    true,
  "context_id":        "witness-3-cosmic-beacon",
  "timestamp_utc":     "<ISO-8601 UTC, second precision>",
  "record_hash":       "<sha256 of canonical_json(all fields above)>"
}
```

record_hash provides field integrity only; it cannot enforce context_id binding (tested in W3-C3(b)).

7. Preregistered Test Cases & Pass/Fail Criteria

W3-C1 — Honest End-to-End Verification

Recompute cosmic_nonce from all three stored witnesses; verify record_hash and every per-witness hash; re-derive CHSH S from stored counts; via the IBM provider API confirm the job id exists and per-setting counts match. **PASS:** all local checks True AND provider job found AND counts match. Provider/network/token unavailability → **GATE-STOP**.

W3-C2 — Tamper Detection (6 sub-trials)

Alter exactly one element per sub-trial and confirm detection:

(a) raw_counts, (b) job_id [nonce + record_hash], (c) calibration, (d) NIST outputValue, (e) astro file_sha256, (f) context_id without record_hash recompute [record_hash].

PASS: all 6 forgeries detected.

W3-C3 — Cross-Context Replay (2 sub-cases)

(a) change context_id without recomputing record_hash → record_hash denies.
 (b) change context_id and recompute record_hash (valid seal — design boundary) → ARK-457 5-dimensional context binding denies (session mismatch). **PASS:** both replays DENIED.

W3-C4 — CHSH Bell-Violation Certification (flagship)

Compute S from stored counts. **PASS iff ALL:** (1) $|S| > 2$; (2) $(|S|-2)/\sigma_S \geq 5.0$; (3) $|S| \leq 2\sqrt{2} + 0.10$. **KILL condition:** if $|S| \leq 2 \rightarrow$ W3-C4 = FAIL, bell_certified = false, published as-is (the fused nonce remains valid as a provenance construction; only the non-classicality certification fails). σ_S from $\text{Var}(E) = (1-E^2)/N$ summed over the four independent correlators.

8. Overall Verdict Rule

PASS iff all four cases (W3-C1..C4) meet their stated criteria and every sub-trial/sub-case within each meets its criterion. Any single failure = overall FAIL, published as-is with full root-cause analysis. GATE-STOP (provider/network/budget) is neither PASS nor FAIL and is logged with reason; the experiment may be re-run once conditions permit, without editing locked code.

9. Related Works

- ARK Corpus (ExecutionProof) concept DOI: 10.5281/zenodo.21398675
 - ARK-457 (cross-context replay; library reused here): DOI 10.5281/zenodo.21421742
 - WITNESS-1: DOI 10.5281/zenodo.21424324 · WITNESS-2: DOI 10.5281/zenodo.21425381
 - WITNESS series concept DOI: 10.5281/zenodo.21424323
 - BELLWETHER series (CHSH Bell-violation entropy witness): prior RF quantum series — WITNESS-3 differs by **fusing** the Bell test with NIST + astrophysical witnesses in a single authorization ProofRecord.
 - NIST Randomness Beacon: NIST IR 8213; <https://beacon.nist.gov/>
 - GW150914: B. P. Abbott et al. (LIGO/Virgo), Phys. Rev. Lett. 116, 061102 (2016); data via GWOSC, <https://gwosc.org/>
-

10. Mandatory Scope Disclaimer

“Results apply within the tested circuit model, backend, qubits, calibration, shot counts, software harnesses, witness sources, and stated experimental conditions. The Bell-violation certification is device-dependent and not loophole-free. The astrophysical data is used only as a public provenance witness, not as a detection or cosmological claim. This is an experimental evidence record, not a universal security proof or production certification.”

Reproduced verbatim in the README and Zenodo deposit description.

11. MANIFEST Lock Reference

witness-3/MANIFEST.sha256 covers this preregistration document and all files under witness-3/src/. It is committed before any QPU job is submitted. Post-lock code changes are prohibited; any pre-execution harness defect triggers the gate-stop protocol and a disclosed, re-hashed fix consistent with WITNESS-1/2 precedent.

Remnant Fieldworks RF Process. Preregistration-first. No post-hoc changes. Soli Deo Gloria.